body, or when repairing any component between the shift lever and the Transmission.

Some questions to ask yourself when performing the road test are as follows:

- Is the complaint or concern what you think the problem is, based on the drivers description of the problem?
- Is the Transmission operating normally, or is there a real problem?
- When does the problem occur?
- Is the problem only in one gear range?
- What temperature does the problem occur?
- Does the vehicle have to sit over night for the problem to occur?
- Does the transmission go into Limp-in mode?

4.3.4 ELECTRONIC PINION FACTOR WARNINGS (IF APPLICABLE)

The pinion factor must be set when replacing the PCM. Note: The pinion factor is a fixed number and cannot be changed or updated in some vehicle applications. If the pinion factor is not set or incorrectly set, any speed related functions will not operate correctly i.e. speedometer, speed control, rolling door locks, other control modules will be affected that depend on speed information.

4.3.5 SERVICE BULLETINS AND RECALLS

Always perform all Safety Recalls and Service Bulletins that are applicable to the problem. Under the provisions of the warranty.

5.0 REQUIRED TOOLS AND EQUIPMENT

DRBIII® (diagnostic read-out box) scan tool - must use the latest release level.
Diagnostic Pinout Box #8815
Evaporative Emissions Leak Detector #8404
Terminal Removal tool #3638
Fuel filler adapter #8382
Fuel pressure adapter (C-6631) or #6539
Fuel pressure kit (C-4799-B) or #5069
Fuel pressure kit #8978
Fuel release hose (C-4799-1)
Jumper wires
Ohmmeter
Oscilloscope
Pressure gauge 0-2068 kPa (0-300 PSI)
Transmission simulator #8833
Electronic Transmission Adapter Kit (Miller # 8333-1A)
Vacuum gauge
Voltmeter

12 volt test light minimum 25 ohms resistance with probe #6801

CAUTION: A 12 volt test light should not be used for the following circuits, damage to the powertrain controller will occur.

- 5-volt Supply
- J1850 PCI Bus
- CCD Bus
- PCI Bus
- CKP Sensor Signal
- CMP Sensor Signal
- Vehicle Speed Sensor Signal
- O2 Sensor Signal

6.0 ACRONYMS

A/C	Air Conditioning
ABS	Anti-lock Brake System
ASD Relay	Auto Shutdown Relay
APPS	Accelerator Pedal Position Sensor
Baro	Barometric Pressure
BCM	Body Control Module
BTS	Battery Temperature Sensor
CAA	Clean Air Act
CAB	Controller Antilock Brakes
CARB	California Air Resources Board
CCD BUS	Chrysler Collision Detection Bus
CKP Sensor	Crankshaft Position Sensor
CKT	Circuit
CMP Sensor	Camshaft Position Sensor
CM840	Cummins Engine controller
CO	Carbon Monoxide
CVI	Clutch Volume Index
DCP Solenoid	Duty-Cycle Purge Solenoid
DLC	Data Link Connector
DRBIII®	Diagnostic Readout Box - 3rd Generation